

1 Maxima of random walks

Exercise 1a (2 points): Draw a path from $(0, 0)$ to $(n, 2r - k)$ that crosses r , and obtain another path by reflecting the path at the last point of crossing r . This gives a path from $(0, 0)$ to (n, k) . Similarly, start with a path from $(0, 0)$ to (n, k) that crosses r and reflect it at the last point of crossing to get a path from $(0, 0)$ to $(n, 2r - k)$.

Exercise 1b (2 points): The probability of a path from $(0, 0)$ to (n, k) while touching or crossing r is the total number of such paths multiplied by the probability of individual paths $1/2^n$. We argued in Exercise 1a that the total number of such paths is equal to the total number of all paths from $(0, 0)$ to $(n, 2r - k)$ which is given by $\binom{n}{\frac{n+2r-k}{2}}$. Hence,

$$\mathbb{P}\{\mathbb{M}_n = 2r - k\} = \binom{n}{\frac{n+2r-k}{2}} \frac{1}{2^n}. \quad (1)$$

Exercise 1c (2 points): The paths with maximum value r are obtained by subtracting the number of paths that exceed $r + 1$ from the number of paths that exceed r . This gives the required probability $p_{n, 2r-k} - p_{n, 2(r+1)-k}$.

Exercise 1d (4 points): As explained in the question, the sum is a telescoping sum:

$$\begin{aligned} \sum_{k=2r-n}^r (p_{n, 2r-k} - p_{n, 2r+2-k}) &= \underbrace{p_{n, n} - p_{n, n+2}}_{k=2r-n} + \underbrace{p_{n, n-1} - p_{n, n+1}}_{k=2r-n+1} + \underbrace{p_{n, n-2} - p_{n, n}}_{k=2r-n+2} \\ &\quad + \underbrace{p_{n, n-3} - p_{n, n-1}}_{k=2r-n+3} + \cdots + \underbrace{p_{n, r+1} - p_{n, r+3}}_{k=r-1} + \underbrace{p_{n, r} - p_{n, r+2}}_{k=r}. \end{aligned} \quad (2)$$

All terms cancel between the successive summands except $p_{n, r}$ in the last summand and $p_{n, r+1}$ in the last but one summand. Since only one of $p_{n, r}$ or $p_{n, r+1}$ is non-zero depending on the parity of n and r , the answer is the appropriate non-zero expression.

2 The first passage time

Exercise 2a (2 points): The probability of paths from $(0, 0)$ to (n, k) having maximum exactly r is $p_{n, 2r-k} - p_{n, 2(r+1)-k}$. Here, replace $n \rightarrow n - 1$, $r \rightarrow r - 1$ and set $k = r - 1$ to get the required probability $p_{n-1, r-1} - p_{n-1, r+1}$.

Exercise 2b (2 points): The probability to reach $\mathbb{M}_n = r$ from $\mathbb{M}_{n-1} = r - 1$ is $\frac{1}{2}$ and this event is independent of the path reaching $r - 1$ at $n - 1$ with maximum $r - 1$. Hence, the

probability $\mathbb{P}\{\mathbb{M}_1 < r, \mathbb{M}_2 < r, \dots, \mathbb{M}_{n-1} < r, \mathbb{M}_n = r\}$ is the product of the probabilities and we get

$$\mathbb{P}\{\mathbb{M}_1 < r, \mathbb{M}_2 < r, \dots, \mathbb{M}_{n-1} < r, \mathbb{M}_n = r\} = \varphi_{r,n} = \frac{1}{2}(p_{n-1,r-1} - p_{n-1,r+1}) . \quad (3)$$

It is a direct calculation to get

$$\varphi_{r,n} = \frac{r}{n} \binom{n}{\frac{n+r}{2}} \frac{1}{2^n} . \quad (4)$$

Exercise 2c (4 points): The event of reaching $(n_1, r-1)$ from $(0,0)$ while visiting $r-1$ for the first time is independent of the event of going from $(n_1, r-1)$ to (n, r) and reaching r for the first time. Further the event of going from $(n_1, r-1)$ to (n, r) is a simple translation of the event of going from $(0,0)$ to $(n-n_1, 1)$, and is governed by the same probabilities. Hence, the probabilities multiply:

$$\begin{aligned} & \mathbb{P}\{\mathbb{M}_1 < r-1, \mathbb{M}_2 < r-1, \dots, \mathbb{M}_{n_1} = r-1, \mathbb{M}_{n_1+1} < r, \dots, \mathbb{M}_{n-1} < r, \mathbb{M}_n = r\} \\ &= \mathbb{P}\{\mathbb{M}_1 < r-1, \mathbb{M}_2 < r-1, \dots, \mathbb{M}_{n_1} = r-1\} \mathbb{P}\{\mathbb{M}_1 < 1, \mathbb{M}_2 < 1, \dots, \mathbb{M}_{n-n_1} = 1\} \\ &= \varphi_{r-1,n_1} \varphi_{1,n-n_1} . \end{aligned} \quad (5)$$

The time instant n_1 can be anywhere between 0 and n , and every choice of n_1 is mutually exclusive from every other choice. Thus, the total probability $\varphi_{r,n}$ is the sum of the above product of probabilities over all possible values of n_1 . This gives

$$\varphi_{r,n} = \sum_{n_1=0}^n \varphi_{r-1,n_1} \varphi_{1,n-n_1} . \quad (6)$$

Exercise 2d (4 points): The product of probability generating functions $\Phi_{r-1}(\alpha)\Phi_1(\alpha)$ is given by

$$\Phi_{r-1}(\alpha)\Phi_1(\alpha) = \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} \alpha^{m+n} \varphi_{r-1,n} \varphi_{1,m} . \quad (7)$$

Writing $j = m + n$, we can rearrange the sum over n and m into a sum over j and n , so that α^j does not depend on the sum over n :

$$\Phi_{r-1}(\alpha)\Phi_1(\alpha) = \sum_{j=0}^{\infty} \sum_{n=0}^j \alpha^j \varphi_{r-1,n} \varphi_{1,j-n} = \sum_{j=0}^{\infty} \alpha^j \sum_{n=0}^j \varphi_{r-1,n} \varphi_{1,j-n} = \sum_{j=0}^{\infty} \varphi_{r,j} . \quad (8)$$

In the last equation, we have used the result derived in Exercise 2c.

Exercise 2e (4 points): We multiply the identity by α^n throughout:

$$\alpha^n \varphi_{1,n} = \alpha^n \sum_{n_1=2}^n q \varphi_{1,n_1-1} \varphi_{1,n-n_1} = q \alpha \sum_{n_1=2}^n \alpha^{n_1-1} \varphi_{1,n_1-1} \alpha^{n-n_1} \varphi_{1,n-n_1} . \quad (9)$$

Summing over n from 2 to ∞ , we get

$$\sum_{n=2}^{\infty} \alpha^n \varphi_{1,n} = q\alpha \sum_{n=2}^{\infty} \sum_{n_1=2}^n \alpha^{n_1-1} \varphi_{1,n_1-1} \alpha^{n-n_1} \varphi_{1,n-n_1} . \quad (10)$$

Defining $m = n - 1$, $m_1 = n_1 - 1$, the right hand side becomes

$$\sum_{n=2}^{\infty} \alpha^n \varphi_{1,n} = q\alpha \sum_{m=1}^{\infty} \sum_{m_1=1}^m \alpha^{m_1} \varphi_{1,m_1} \alpha^{m-m_1} \varphi_{1,m-m_1} . \quad (11)$$

Further defining $m_2 = m - m_1$, we get (this is the reverse of the step in Exercise 2d)

$$\sum_{n=2}^{\infty} \alpha^n \varphi_{1,n} = q\alpha \sum_{m_1=1}^{\infty} \sum_{m_2=1}^{\infty} \alpha^{m_1} \varphi_{1,m_1} \alpha^{m_2} \varphi_{1,m_2} . \quad (12)$$

Now, we use $\Phi_1(\alpha) = \sum_{n=1}^{\infty} \alpha^n \varphi_{1,n} = \alpha p + \sum_{n=2}^{\infty} \alpha^n \varphi_{1,n}$, we get

$$\Phi_1(\alpha) = \alpha p + q\alpha (\Phi_1(\alpha))^2 . \quad (13)$$

Solving the quadratic equation for $\Phi_1(\alpha)$, we get

$$\Phi_1(\alpha) = \frac{1 - \sqrt{1 - 4pq\alpha^2}}{2q\alpha} . \quad (14)$$

Exercise 2f (4 points):

The sum

$$\sum_{n=r}^{\infty} \varphi_{r,n} = \sum_{n=r}^{\infty} \frac{r}{n} \binom{n}{\frac{n+r}{2}} \frac{1}{2^n} , \quad (15)$$

is obtained by either direct summation, or more economically, taking the limit $\alpha \rightarrow 1$ of the generating function $\Phi_r(\alpha)$.

We have

$$\lim_{\alpha \rightarrow 1} (\Phi_1(\alpha))^r = 1 . \quad (16)$$

This shows that the symmetric random walk passes through r at some time instant (which instant is immaterial) given sufficient time.

3 The normal approximation to the binomial distribution

Exercise 3a (4 points): We have $n_H = r + \nu$, $n = 2\nu$ and $p = \frac{1}{2}$, giving

$$a_r = \binom{2\nu}{\nu + r} \frac{1}{2^{2\nu}} . \quad (17)$$

Expanding the binomial coefficient, we have

$$a_r = \frac{(2\nu)!}{(\nu+r)!(\nu-r)!} \frac{1}{2^{2\nu}} = \frac{(\nu-r+1)}{(\nu+r)} \frac{(2\nu)!}{(\nu+r-1)!(\nu-r+1)!} \frac{1}{2^{2\nu}} = \frac{(\nu-r+1)}{(\nu+r)} a_{r-1} . \quad (18)$$

Applying the recursion successively to a_{r-1} , a_{r-2} and so on till a_0 , we get

$$a_r = \frac{(\nu-r+1)}{(\nu+r)} \frac{(\nu-r+2)}{(\nu+r-1)} \cdots \frac{(\nu-1)}{(\nu+2)} \frac{\nu}{(\nu+1)} a_0 . \quad (19)$$

Exercise 3b (4 points): We have

$$a_r = a_0 \frac{(1 - \frac{r-1}{\nu})(1 - \frac{r-2}{\nu}) \cdots (1 - \frac{1}{\nu})}{(1 + \frac{r}{\nu})(1 + \frac{r-1}{\nu}) \cdots (1 + \frac{2}{\nu})(1 + \frac{1}{\nu})} \approx a_0 \frac{e^{-\frac{r-1}{\nu}} e^{-\frac{r-2}{\nu}} \cdots e^{-\frac{1}{\nu}}}{e^{\frac{r}{\nu}} e^{\frac{r-1}{\nu}} \cdots e^{\frac{1}{\nu}}} . \quad (20)$$

In the last expression, note that every exponent except $e^{r/\nu}$ appears twice, giving

$$a_0 \frac{e^{-\frac{r-1}{\nu}} e^{-\frac{r-2}{\nu}} \cdots e^{-\frac{1}{\nu}}}{e^{\frac{r}{\nu}} e^{\frac{r-1}{\nu}} \cdots e^{\frac{1}{\nu}}} = a_0 \exp \left(-\frac{2}{\nu} [1 + 2 + \cdots + (r-1)] - \frac{r}{\nu} \right) . \quad (21)$$

Using $\sum_{i=1}^{r-1} i = \frac{1}{2}r(r-1)$, we get

$$a_r \approx a_0 e^{-r^2/\nu} . \quad (22)$$

Exercise 3c (2 points): We get, using Stirling's formula,

$$a_0 = \frac{(2\nu)!}{(\nu)!(\nu)!} \frac{1}{2^{2\nu}} \approx \frac{\sqrt{2\pi}(2\nu)^{2\nu+\frac{1}{2}} e^{-2\nu}}{\left(\sqrt{2\pi}\nu^{\nu+\frac{1}{2}} e^{-\nu}\right)^2} \frac{1}{2^{2\nu}} . \quad (23)$$

Most of the factors in the numerator and denominator cancel, giving the desired result:

$$a_0 \approx \frac{1}{\sqrt{\pi\nu}} . \quad (24)$$

This implies that

$$a_r \approx \frac{1}{\sqrt{\pi\nu}} e^{-r^2/\nu} . \quad (25)$$

Since $k = n_H - n_T = 2n_H - n = 2r$, we have

$$p_{2\nu,k} \approx \frac{1}{2\sqrt{\pi\nu}} e^{-k^2/4\nu} , \quad \text{that is} \quad p_{n,k} = \frac{1}{\sqrt{2\pi n}} e^{-k^2/2n} , \quad (26)$$

which is a normal distribution with mean 0 and variance n . The extra factor of 1/2 in the first approximation comes from the fact that $k = 2r$. The probabilities must be treated as densities which change by a factor of 2 when going from the variable k to the variable r .

It is not too difficult to repeat this derivation for the asymmetric random walk with probabilities p, q for heads and tails respectively, with the result that

$$p_{n,k} \approx \frac{1}{\sqrt{8\pi p q n}} e^{-k^2/8p q n} . \quad (27)$$

which is a normal distribution with mean 0 and variance $4pqn$.

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